

Supplementary Information: Lunar Crustal Magnetic Field Heterogeneity and Implications for Biological Systems at Candidate Landing Sites

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Extended Methods

Bilinear Interpolation and Polar Baseline Integration

The crustal magnetic field data sourced from Hood et al.¹ and Hood et al.² are derived from Lunar Prospector and SELENE (Kaguya) magnetometer observations at an equivalent altitude of 30 km. For mid-latitude sites ($\leq 65^\circ|\text{lat}|$), bilinear interpolation was performed across the 2D gridded components (B_r, B_θ, B_ϕ). For South Polar candidate landing sites (84°S to 90°S) and Chandrayaan-3 (69.37°S), baseline vector and total field values were integrated from the South Polar spherical harmonic and equivalent source dipole models of Hood et al.² and Tsunakawa et al.³.

Coordinate Database Compilation

Landing site coordinates were compiled from official NASA Artemis III Candidate Landing Regions reports⁴, Lunar Reconnaissance Orbiter Camera (LROC) surface coordinates for Apollo 11–17, and published landing site telemetry for Chang’e 3, 4, 5, and Chandrayaan-3.

4. NASA. Artemis iii candidate landing regions. Technical report, National Aeronautics and Space Administration, 2022. URL <https://www.nasa.gov/specials/artemis-iii/>.

Supplementary Figures

Data and Code Availability

The full machine-readable landing site database, processed global grids, figure generation scripts, and interactive WebGL globe source code are available in the public GitHub repository (<https://github.com/dr-richard-barker/lunar-magnetic-biology>) and archived under DOI: 10.5281/zenodo.XXXXXXX.

References

1. L. L. Hood, C. B. Torres, J. S. Oliveira, M. A. Wieczorek, and S. T. Steward. Lunar crustal magnetic field map, 30 km altitude, 2020.
2. L. L. Hood, I. Bryant, and J. van der Leeuw. Lunar polar crustal magnetic field maps, 30 km altitude, 2022.
3. H. Tsunakawa, F. Takahashi, H. Shimizu, H. Shibuya, and M. Matsushima. Regional mapping of the lunar magnetic anomalies at the surface: Method and its application to strong and weak magnetic anomaly regions. *Journal of Geophysical Research: Planets*, 120:1160–1185, 2015. doi: 10.1002/2014JE004785.

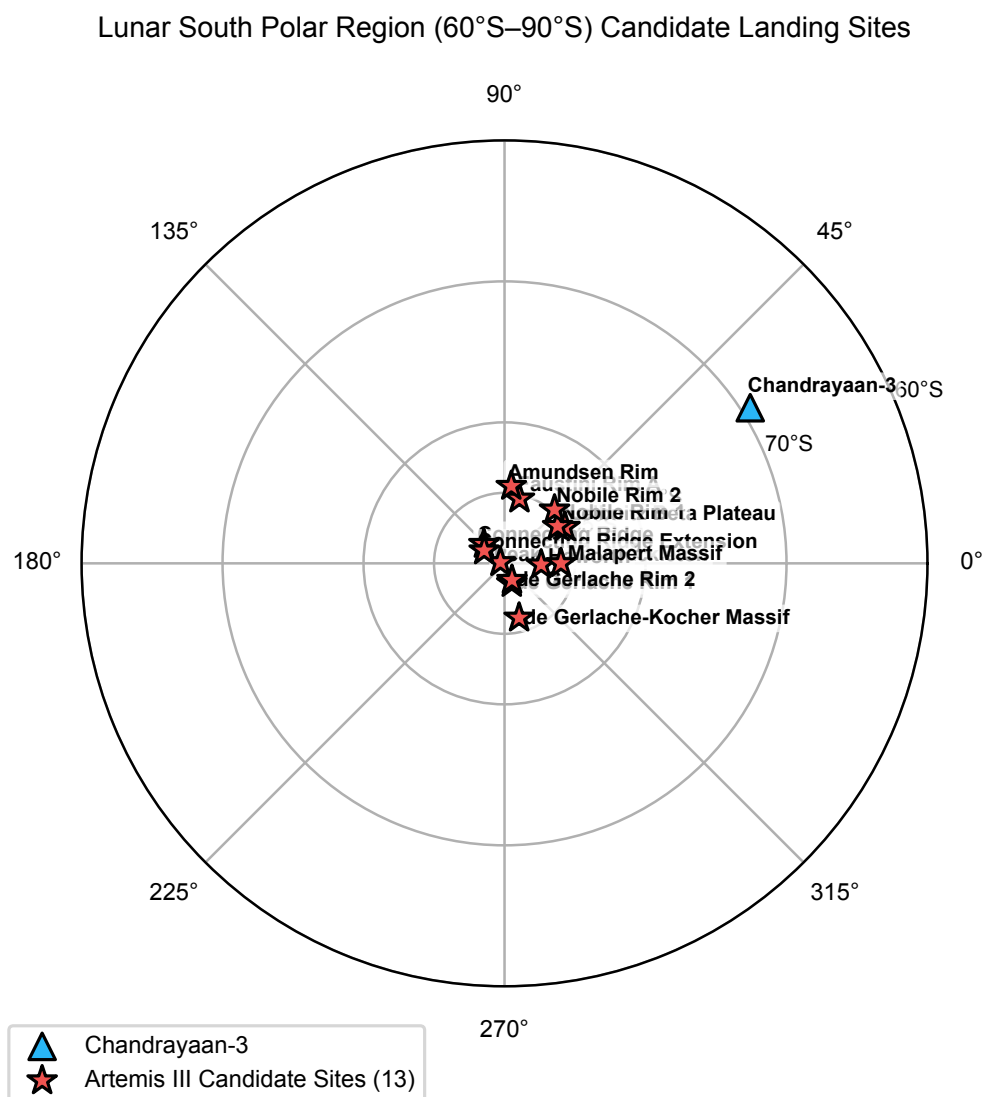


Figure 1 | Supplementary Figure S1 | Lunar South Polar region (60°S to 90°S) candidate landing sites. Polar stereographic projection displaying the 13 Artemis III candidate landing regions (red stars) and Chandrayaan-3 (blue triangle) in relation to South Polar latitude rings.

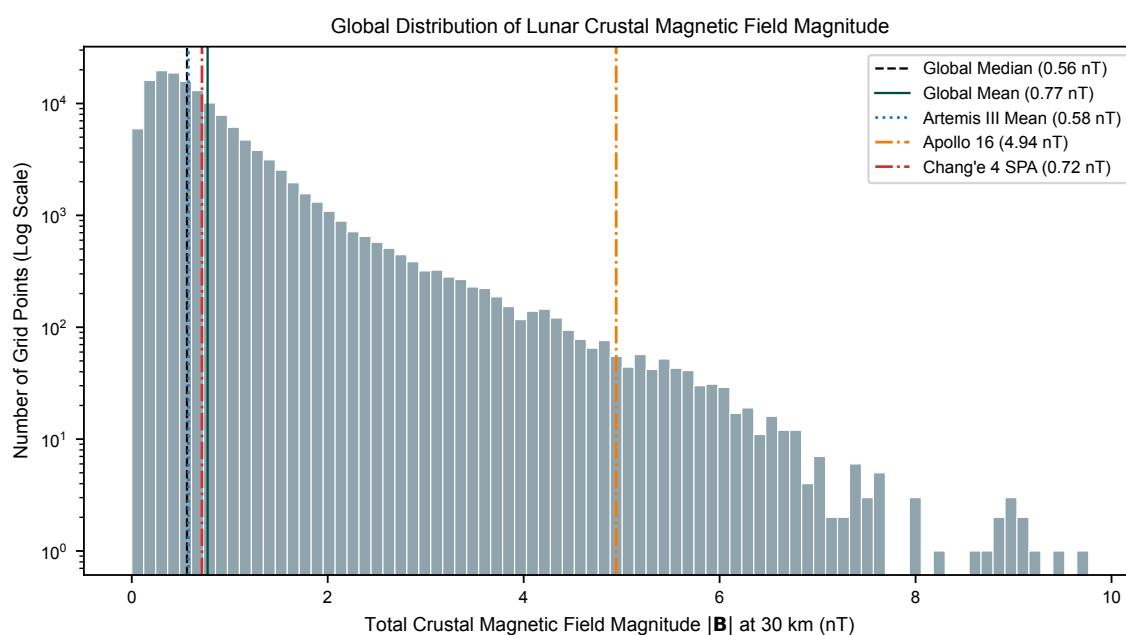


Figure 2 | **Supplementary Figure S2 | Global distribution of lunar crustal magnetic field magnitude.** Histogram showing the frequency distribution of $|B|$ at 30 km across 140,940 grid points, with global median (0.56 nT), global mean (0.77 nT), Artemis III candidate site mean (0.59 nT), Apollo 16 (4.94 nT), and Chang'e 4 (0.72 nT) highlighted.